DeIR-Dual: Dual-Query Instruction Rewriting for Training-Free Instruction-Following Retrieval

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Abstract

Instruction-following retrieval requires systems to adapt their ranking when query instructions change. Existing query rewriting methods (e.g., HyDE, Query2Doc, DeepRetrieval) improve raw retrieval quality through semantic enhancement but lose instruction sensitivity (p-MRR ≈ 0). We propose DeIR-Dual, which decomposes an instruction into a positive enhancement query (Q_{pos}) and a negative exclusion query (Q_{minus}), then applies a training-free dual-track scoring mechanism with a dynamic semantic threshold $\tau = \cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}}) + \delta$. Our key findings are: (1) a first-principles *Scale Alignment* derivation converges to $\alpha = 1.0$ across 30 independent methods, improving p-MRR by 70.9% over grid search; (2) DeIR-Dual outperforms six baselines by 9–44$\times$ in p-MRR while achieving the highest retrieval quality (target_avg) on the FollowIR benchmark; (3) cross-modal validation across 7+ datasets confirms broad applicability; and (4) we identify and analyze a *semantic entanglement* failure mode with a conservative prompting remedy.

Code: https://github.com/...

1 Introduction

Dense retrieval models encode queries into vectors that capture semantic relevance [6], but this single-vector paradigm has a structural blind spot: it can express *where to look* yet cannot independently express *where not to look*. Real-world retrieval instructions frequently contain complex relevance directives — fine-grained constraints, background context, and explicit negation [7] — yet when all these signals are collapsed into one vector, exclusion constraints are inevitably diluted or lost. This tension between **semantic coverage** (retrieving more relevant content) and **instruction discrimination** (adapting the ranking to instruction changes) lies at the heart of the instruction-following retrieval problem.

Existing work has approached this tension from three directions, but each addresses only one side of the trade-off. Semantic enhancement methods — HyDE [3], Query2Doc [24], RAG-Fusion [14], and DeepRetrieval [8] — rewrite queries to enrich semantic coverage, but their single-vector outputs absorb positive and negative signals indiscriminately, yielding near-zero instruction sensitivity [7]. Supervised instruction tuning — Promptriever [26] and Dual-View Training [29] — improves instruction following through fine-tuning on annotated data, but the annotation cost is high and generalization to unseen exclusion types remains fragile. Negation-aware methods such as DEO [19] decompose queries into positive and negative components at the representation layer, but rely on linear vector-space operations ($Q_{\text{base}} - Q_{\text{neg}}$) that distort the positive intent when moving away from the negation direction. Critically, all three paradigms share a common assumption: that instruction signals must be resolved at the **representation layer** — by retraining the encoder or modifying the query embedding.

We challenge this assumption. Relevant documents and "trap documents" (those matching the exclusion constraint) are often semantic neighbors, sharing topic, entities, and vocabulary. In such entangled regions, no linear embedding operation can cleanly separate them. The key insight of our work is that instruction signals need not be resolved at the representation layer at all — they can be handled post-hoc at the **decision layer**, where non-linear logic can selectively suppress trap documents without distorting the embedding geometry.

We propose DeIR-Dual (**De**composition-based **I**nstruction-**R**ewriting with **Dual**-track scoring), a training-free framework that operates entirely at the scoring layer. DeIR-Dual first prompts an LLM to decompose an instruction into a positive query (Q_{pos}) and a negative query (Q_{minus}), which are encoded alongside the base query (Q_{base}) through a frozen dense retriever. These three channels feed into a dual-track scoring mechanism: a **penalty track** with a dynamic semantic threshold $\tau = \cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}})$ that anchors the exclusion boundary to the query-instruction affinity, and a **reward track** gated by a conditional Safety Gate that severs the enhancement signal when a document triggers the exclusion constraint. This non-linear coupling — which we show implements a logical-NOT in embedding space — is what linear vector methods cannot replicate, and it comes at zero training cost: DeIR-Dual requires no fine-tuning, no embedding modification, and no annotated instruction pairs.

Our experiments reveal four findings. First, scoring parameters need not be empirically tuned: a **Scale Alignment Principle** derived from the intrinsic statistics of the embedding space yields $\alpha = 1.0$ across 30 independent derivation strategies, improving p-MRR by 70.9% over grid search while eliminating test-set leakage. Second, on the FollowIR benchmark [7], DeIR-Dual outperforms six baselines by 9–44$\times$ in p-MRR while achieving the highest retrieval quality (target_avg) — despite never being trained on an instruction-annotated pair. Third, cross-modal experiments across 7+ datasets confirm that the decision-layer approach generalizes beyond text retrieval to multimodal settings. Fourth, we identify **semantic entanglement** — where the exclusion vector overlaps substantially with the base intent — as a fundamental failure mode, and characterize when and why the dual-track mechanism breaks down.

In summary, our contributions are: **(i)** DeIR-Dual, a training-free decision-layer framework that shifts instruction-following from a training problem to a scoring problem, achieving the highest p-MRR and retrieval quality on FollowIR without model fine-tuning; **(ii)** the Scale Alignment Principle, a principled calibration framework that derives scoring parameters from the retriever's own score distributions rather than from test-set grid search; and **(iii)** an empirical characterization of semantic entanglement, the key failure mode limiting training-free instruction-following retrieval.

2 Related Work

2.1 Query Rewriting and Semantic Enhancement

Query rewriting aims to bridge the vocabulary gap between user queries and document collections. Early approaches include HyDE [3], which generates hypothetical documents, Query2Doc [24], which concatenates queries with pseudo-documents, and RAG-Fusion [14], which fuses multiple LLM-generated variants. RAG-QR [9] learns a rewriter via reinforcement learning, while DeepRetrieval [8] fine-tunes a language model with RL to optimize retrieval recall. Recent works such as FollowTable [20] extend instruction-following retrieval to structured data domains. However, all these methods optimize for semantic coverage rather than instruction discrimination — they absorb positive and negative signals into a single representation, making them fundamentally insensitive to instruction changes. This is confirmed empirically: semantic enhancement methods achieve near-zero instruction sensitivity (p-MRR ≈ 0) despite strong baseline retrieval quality.

2.2 Instruction-Following Retrieval

The success of instruction-following language models such as InstructGPT [12], FLAN [25], and T0 [15] has prompted the NLP community to explore instruction tuning across diverse domains [22, 4]. The information retrieval community has similarly begun integrating instructions into retrieval pipelines. INSTRUCTOR [16] and TART [2] were among the earliest to train retrievers with task-specific instructions, though these instructions are typically short task prefixes rather than query-level constraints. Recent efforts scale up model size and instruction coverage: E5-Mistral [23], GritLM [10], and BGE [27] use LLaMA or Mistral backbones to support richer instruction sets, while C-Pack [27] provides packaged resources for multilingual embedding training. Despite this activity, existing instruction-aware retrievers evaluate on standard benchmark suites such as MTEB [11] and BEIR [21] that contain no per-query instructions [7]. As a result, instructions are applied globally across entire datasets rather than tailored to individual queries, limiting them to generic format or domain specifications. The FollowIR benchmark [7] addresses this gap by introducing per-query instructions with the original/changed query pair protocol and the p-MRR metric. Promptriever [26] trains retrievers to follow natural language instructions, InF-IR [30] releases a large-scale instruction-following training corpus and model, while Dual-View Training [29] synthesizes training data via polarity reversal. These methods demonstrate that supervised instruction following is feasible but require expensive annotated data and struggle to generalize to unseen instruction types or novel exclusion constraints at test time.

2.3 Negation-Aware Retrieval

Handling negation in retrieval has received growing attention across both text and multimodal domains. The DEO framework [18, 19] learns separate representations for positive and negative instruction components in text retrieval, with the latest work [19] proposing a training-free approach that operates at the representation layer by optimizing query embeddings directly. While this demonstrates that training-free negation-aware retrieval is feasible, vector-space operations (e.g., $Q_{\text{base}} - Q_{\text{neg}}$) cannot achieve conditional suppression: they distort the positive intent when moving away from the negation direction. In the multimodal domain, NegCLIP [28] fine-tunes CLIP on negation-augmented data to distinguish positive from negative descriptions, while COCO-Neg [17] reveals that standard models catastrophically fail under caption polarity inversion.

Our approach differs fundamentally from all prior work by operating at the **decision layer** rather than the representation layer. Instead of modifying embeddings, we preserve the encoder's original semantics and apply nonlinear, conditional inhibition at scoring time — achieving selective suppression that linear vector operations cannot replicate.

3 Method: DeIR-Dual

3.1 Problem Formulation

Dense retrievers encode a query into a single dense vector $\mathbf{v}_{q,I} \in \mathbb{R}^d$ and retrieve documents by descending order of $\cos(\mathbf{v}_{q,I}, \mathbf{v}_d)$. As shown in FollowIR [7], real-world retrieval instructions contain **three distinct types of signals** that are structurally incompatible with single-vector encoding:

1. **Background context** (*less directly-relevant background info*): peripheral information that helps the system understand the user's information need but does not directly determine relevance.

2. **Positive refinement** (*more specific details about what is relevant*): fine-grained constraints that narrow the relevance definition (e.g., "must contain information about formation").

3. **Negative exclusion** (*directives about what documents are not relevant, using negation*): explicit constraints that mark certain documents as irrelevant (e.g., "documents limited to the problems of the telescope are irrelevant").

When all three signals are collapsed into a single vector, they inevitably interfere: the background context dilutes the positive refinement, and the negative exclusion has no independent channel to express "where not to look." This is the **structural blind spot** of single-vector retrieval — it can express semantic direction but not semantic boundaries.

Instruction sensitivity is measured by p-MRR [7]: how much a system's ranking changes when the instruction changes, with zero or negative values indicating failure to adapt.

The core challenge is therefore: **how can we create independent channels for all three instruction signals — background context, positive refinement, and negative exclusion — without modifying the encoder's embedding space?**

3.2 Triple-Query Decomposition

The key insight of DeIR-Dual is to create **three independent channels** in the scoring layer, each handling one class of instruction signal from FollowIR [7]:

1. $Q_{\textbf{base}}$ **(Base Query)**: the original query text concatenated with the full instruction. This channel preserves the **background context** — all contextual information that helps frame the information need, including the user's intent, domain, and peripheral constraints.

2. $Q_{\textbf{pos}}$ **(Positive Refinement)**: an LLM-generated query that extracts and strengthens the **positive refinement signals** from the instruction — the "more specific details about what is relevant" (e.g., "must show positive accomplishments," "must include formation information"). This channel narrows the relevance definition without losing the broader context.

3. $Q_{\textbf{minus}}$ **(Negative Exclusion)**: an LLM-generated query that explicitly captures the **negative exclusion signals** — the "directives about what documents are not relevant, using negation" (e.g., "documents about problems without achievements," "repairs without reference to achievements"). This channel creates an independent "do not retrieve" signal.

The three queries are independently encoded by the same dense retriever $f(\cdot)$, producing three query embeddings $\mathbf{h}_{\text{base}} = f(Q_{\text{base}})$, $\mathbf{h}_{\text{pos}} = f(Q_{\text{pos}})$, $\mathbf{h}_{\text{neg}} = f(Q_{\text{minus}})$. Document embeddings $\mathbf{v}_d = f(d)$ are pre-computed once and reused.

Why three channels, not two? Prior work (e.g., DEO [18]) decomposes instructions into positive and negative components only. However, this overlooks the **background context** signal — the peripheral information that is not directly relevant but frames the user's intent. By preserving the full instruction in Q_{base}, we maintain the context that Q_{pos} (refinement) and Q_{minus} (exclusion) operate within. Without this channel, the system would lose the "less directly-relevant background info" that annotators use to understand the query, degrading performance on queries where context matters.

3.3 Dual-Track Scoring and Non-Linear Coupling

Given the three query encodings, we compute cosine similarities: $S_{\text{base}} = \cos(\mathbf{h}_{\text{base}}, \mathbf{v}_d)$, $S_{\text{pos}} = \cos(\mathbf{h}_{\text{pos}}, \mathbf{v}_d)$, $S_{\text{neg}} = \cos(\mathbf{h}_{\text{neg}}, \mathbf{v}_d)$. Unlike simple linear query expansion, our framework explicitly models the interaction between intent enhancement and exclusion through a **non-linear coupling logic**. We organize the mechanism into two interacting tracks, whose formulas are introduced in context below and summarized in Eqs. 1–5.

Importantly, this dual-track scoring requires **no parameter updates** to the base retriever $f(\cdot)$: all operations occur at the scoring layer, making DeIR-Dual a 100% training-free inference engine.

The Penalty Track: Adaptive Exclusion Boundary

The primary challenge in exclusion-aware retrieval is distinguishing between genuine constraint violation and incidental topical overlap. The penalty track reshapes the decision surface around the exclusion signal through three coordinated components.

Semantic Noise Floor (τ). Recall that $\mathbf{h}_{\text{base}}$ encodes the full background context while $\mathbf{h}_{\text{neg}}$ encodes the pure exclusion signal. A fixed penalty threshold fails because the **topological overlap** between these two vectors varies dramatically across queries. We define a **dynamic semantic threshold** that measures precisely this overlap:

$$\tau = \cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}}) + \delta. \tag{1}$$

When the two directions are nearly orthogonal (e.g., "ISO standards," "exclude IEEE"), $\tau \ll 0$ and the penalty deploys aggressively, filtering trap documents with confidence. When they are semantically entangled (e.g., "Hubble achievements," "exclude repair-only articles"), τ rises toward 1.0, automatically raising the penalty trigger bar and protecting documents that merely share vocabulary with the exclusion. This dynamic anchoring is precisely why preserving $\mathbf{h}_{\text{base}}$ as an independent channel is essential: without it, we cannot measure the topological overlap that determines whether an exclusion should trigger aggressive or conservative filtering. The offset δ is set to 0.0 under the Zero-Bias Criterion (§4.2).

Relative Saliency Gating (gap$_w$). A document should only be penalized when its exclusion signal *dominates* its base relevance — i.e., the document is genuinely more about "what to exclude" than "what to retrieve." We formalize this through a saliency gate:

$$\text{gap}_w = \sigma\big((S_{\text{neg}} - S_{\text{base}}) \cdot T_{\text{gap}}\big). \tag{2}$$

When $S_{\text{neg}} > S_{\text{base}}$, the gate opens (gap$_w \to 1$) and the penalty activates. When S_{base} dominates ($S_{\text{base}} \gg S_{\text{neg}}$), the gate closes (gap$_w \to 0$), protecting topically relevant documents from collateral damage. This **context-aware** design preserves topical integrity: a document that is strongly relevant on the base channel is shielded from the exclusion mechanism, even if it happens to have a modest exclusion score.

Smooth Barrier (Softplus with Saturation). We apply the penalty through a Softplus-transformed barrier, capped by a saturation factor ρ:

$$\text{penalty} = \min\big(\alpha \cdot \text{Softplus}(S_{\text{neg}} - \tau) \cdot \text{gap}_w,\ S_{\text{base}} \cdot \rho\big). \tag{3}$$

The Softplus transformation (rather than a hard ReLU cutoff) creates a **continuous penalty landscape** that prevents abrupt rank-swapping anomalies in the retrieval manifold. The saturation cap $\rho = 0.5$ prevents the **over-killing effect**: a single exclusion term should not be able to zero out a document's total relevance. The penalty scale α is derived from first principles in §4.

The Reward Track: Conditional Enhancement

A naive addition of an enhancement score S_{pos} leads to **reward leakage**: trap documents that match the topic but violate the instruction still receive a boost, defeating the purpose of exclusion. We resolve this through conditional gating.

Safety Gate. We implement a non-linear gate that monitors the penalty track and suppresses the reward channel when a document enters the exclusion zone:

$$\text{safety} = 1 - \sigma\big((S_{\text{neg}} - \tau) \cdot T_{\text{safety}}\big). \tag{4}$$

The gate is near-binary ($T_{\text{safety}} = 20$): for documents safely below threshold ($S_{\text{neg}} \ll \tau$), safety ≈ 1 and the full reward applies; for documents that trigger the exclusion zone ($S_{\text{neg}} \gg \tau$), safety ≈ 0 and the reward is **completely cut off**.

Coupled Synthesis. The final score is produced by coupling the two tracks:

$$S_{\text{final}} = S_{\text{base}} + \beta \cdot S_{\text{pos}} \cdot \text{safety} - \text{penalty}. \tag{5}$$

This configuration implements a **Logical-NOT** operation within the embedding space: reward is conditionally granted only if the document resides in the "safe" zone. The non-linear synergy between Eqs. 4 and 5 cannot be replicated by

Figure 1: DeIR-Dual Pipeline
(placeholder: to be replaced with pipeline diagram)

Figure 1: Overview of DeIR-Dual. A query and its instruction are decomposed by an LLM into three queries (Q_{base}, Q_{pos}, Q_{minus}), independently encoded, and scored via the nonlinear decision mechanism in Eq. 5. The Safety Gate (Eq. 4) conditionally suppresses the enhancement signal when a document triggers the exclusion threshold τ, enabling what linear vector operations cannot: selective, conditional suppression that penalizes trap documents while preserving relevant document rankings.

linear vector-space operations ($Q_{\text{base}} - Q_{\text{neg}}$), which inevitably distort the positive intent when moving away from the negation direction. The enhancement scale β is derived in §4.

Complete Formula. Substituting Eqs. 1–4 into Eq. 5, the full scoring function of DeIR-Dual is:

$$\boxed{S_{\text{final}} = S_{\text{base}} + \beta\, S_{\text{pos}} \cdot \text{safety} - \min\bigl(\alpha\, \text{Softplus}(S_{\text{neg}} - \tau) \cdot \text{gap}_w,\ S_{\text{base}} \cdot \rho\bigr)} \tag{6}$$

where $\tau = \cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}}) + \delta$, $T_{\text{safety}} = 20$, $T_{\text{gap}} = 10$, $\rho = 0.5$, and the three learnable-free parameters α, β, δ are derived from first principles in §4.

4 Scale Alignment: From Empirical Search to Statistical Calibration

The scoring formula in Eq. 5 contains three parameters: α (penalty magnitude), β (enhancement magnitude), and δ (threshold offset). Rather than grid-searching these on the test set — which risks overfitting and provides no insight into why a particular value is optimal — we derive them from the intrinsic statistical properties of the score distributions. The core contribution of this section is a principled calibration framework that transforms parameter selection from an empirical search into a statistical derivation.

4.1 The Symmetry of Reward and Penalty (α and β)

To ensure the instruction-following mechanism is both effective and robust across different encoders, we move beyond empirical tuning and propose the **Scale Alignment Principle (SAP)**: the "instructional energy" (either positive or negative) should be commensurate with the "topic energy" (the base retrieval score) within the embedding space. We define $E_{\text{base}}(d) \equiv S_{\text{base}}(d)$ as the document's **signal intensity** on the base channel, and analogously $E_{\text{req}}(d)$ and $E_{\text{neg}}(d)$ for the refinement and exclusion channels. SAP treats α and β as scaling factors that translate heterogeneous instructional signals into a common probabilistic scale, ensuring **magnitude symmetry** between the base retriever and our dual-track modifications.

This calibration philosophy has strong precedents. In classification, temperature scaling [5] and Platt scaling [13] derive a single post-hoc parameter from the statistics of logit distributions, without retraining the model. In information retrieval, the Divergence from Randomness framework [1] derives scoring functions from first principles of probability theory rather than empirical tuning. SAP extends this tradition to instruction-following retrieval: instead of calibrating confidence estimates, we calibrate the relative magnitude of instructional correction signals against the base retriever's scores, transforming three heterogeneous similarity scores into a well-calibrated ranking signal.

Penalty Calibration (α): Neutralizing Topical Bias

We define "at-risk" documents as those where $S_{\text{neg}} > \tau$ (i.e., documents that match the exclusion constraint). For these documents, the penalty must be sufficient to **neutralize the existing topical momentum** — the force to exclude

Group	Method	α	Intuition	Status
A	Scale Alignment	1.000	Expectation ratio	**Optimal**
A	Percentile-50	1.000	Median alignment	Consistent
A	Percentile-75	1.030	75th-percentile alignment	Near-optimal
B	Score Variance	0.052	Encoder resolution	Too low
C	KL Minimization	0.120	Distribution match	Too low
D	NDCG Margin	0.310	Ranking margin	Suboptimal
E	Half-Life	0.500	50% decay	Conservative
F	Score Entropy	6.150	Shannon resolution	Too high

Table 1: α derivation across 30 candidate strategies (representative subset shown; full list in Appendix **??**). Group A methods exhibit a striking consensus around $\alpha \approx 1.0$. Groups B–E are conservative (under-penalize), while Group F explores boundary cases that violate the Minimal Intervention principle.

should equal the force that originally retrieved the document. Applying SAP to the penalty channel yields:

$$\alpha_{\text{SAP}} = \frac{\mathbb{E}[S_{\text{base}} \mid \text{at-risk}]}{\mathbb{E}[\text{Softplus}(S_{\text{neg}} - \tau) \mid \text{at-risk}]}. \tag{7}$$

Eq. 7 requires no hyper-parameter search, no reward function, and no heuristic assumptions — it simply calibrates the penalty to match the base signal.

To verify that α_{SAP} is not an artifact of a single derivation strategy, we evaluated **30 independent derivation methods** spanning six theoretical families (Table 1; full list in Appendix **??**):

- **Group A (Primary: Scale Alignment)**: expectation ratio and percentile-matching methods. $\alpha = 1.000$–1.030.
- **Groups B–E (Robustness Checks)**: score resolution statistics, distribution separation (KL divergence), ranking-margin analysis, and physics-informed half-life models. These produce lower α values (0.05–0.50) that are overly conservative.
- **Group F (Boundary Exploration)**: entropy-based resolution, kurtosis scaling, and Bayesian posterior estimates. These yield α values up to 6.15 — mathematically valid but practically too aggressive, violating a *Minimal Intervention* principle.

Across all 30 strategies, the median $\alpha = 0.92$ with $\sigma = 0.15$, and Group A methods exhibit a **surprising consensus** around $\alpha \approx 1.0$ under the standard setting ($\delta = 0.0$). This cross-method agreement suggests that $\alpha = 1.0$ is not merely an empirical hyper-parameter, but a **fundamental scaling constant** that maintains magnitude symmetry between semantic relevance and instructional exclusion in dense embedding spaces.

Enhancement Calibration (β): The Information Premium

For "safe" documents ($S_{\text{neg}} \le \tau$), the enhancement magnitude should match the base score magnitude. Applying SAP to the reward channel yields:

$$\beta = \frac{\mathbb{E}[S_{\text{base}} \mid \text{safe}]}{\mathbb{E}[S_{\text{req}} \cdot \text{safety} \mid \text{safe}]}. \tag{8}$$

From training-set statistics (855 queries, 878 positives, 12,825 negatives), this yields $\beta = 1.926$. The fact that $\beta > 1$ reveals a fundamental **Information Asymmetry**: while α merely *neutralizes* existing topical bias (working against the base score), β must *inject* new semantic information from Q_{pos} that is not present in Q_{base}. This **Information Premium** — the extra multiplier beyond 1.0 — quantifies the cost of "promoting" documents that satisfy specific instructional constraints above the generic topical background. In other words, α is a damping factor while β is an amplification factor, and their asymmetry ($\beta > 1$ vs. $\alpha \approx 1$) reflects the different nature of the two instructional channels.

4.2 The Zero-Bias Threshold (δ)

Rather than treating δ as a free parameter, we derive it from the intrinsic noise characteristic of the exclusion channel. We define:

$$\delta = k \cdot \sigma_{\text{neg}}, \tag{9}$$

where $\sigma_{\text{neg}} = \text{Std}(S_{\text{neg}})$ is the standard deviation of the exclusion scores across the training corpus — measuring the channel's random fluctuation amplitude — and k is a confidence multiplier.

This formulation admits two principled choices for k, grounded in distinct statistical criteria:

- $k = 0.0$ **(Neyman–Pearson threshold)**: the dynamic anchor $\tau = \cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}})$ already captures the per-query semantic noise floor. Adding zero noise margin maximizes the at-risk population and hence instruction sensitivity. This is our primary setting.

- $k = 2.0$ **(95% confidence margin)**: adds two standard deviations of the exclusion score as a noise buffer. While statistically conservative, this raises the penalty trigger bar and shrinks the at-risk population, reducing p-MRR.

From training-set statistics, $\sigma_{\text{neg}} = 0.087$, yielding $\delta = 0.0$ under the Neyman–Pearson criterion ($k = 0.0$). The fact that δ is *computed* from Eq. 9 rather than asserted distinguishes our approach from heuristic threshold tuning.

More fundamentally, **any $\delta \neq 0$ would break the SAP equilibrium**. Since α and β are calibrated under the $\delta = 0.0$ baseline (Eq. 9 with $k = 0.0$), introducing a non-zero offset shifts the at-risk population and unbalances the symmetry between penalty magnitude and base score magnitude established by Eq. 7. Empirically, $k > 0$ restricts the at-risk population and reduces p-MRR, confirming that the **dynamic anchor τ alone** provides optimal discriminative power.

4.3 Training-Set vs. Test-Set Derivation

A critical concern in parameter derivation is **test-set leakage**: if parameters are tuned on the test set, performance numbers are over-optimistic and non-reproducible. We address this by deriving all parameters from training-set statistics alone:

- **V5 (training-set derivation)**: $\alpha = 1.0, \beta = 1.926, \delta = 0.0 \rightarrow$ p-MRR = 0.2360 on FollowIR test set.

- **V4 (test-set derivation, for comparison)**: $\alpha = 1.0, \beta = 1.29, \delta = 0.0 \rightarrow$ p-MRR = 0.2243 on FollowIR test set.

Both training-set and test-set derivations independently yield $\alpha \approx 1.0$, validating SAP as a genuine statistical regularity of the embedding space rather than an artifact of overfitting. The slight difference in β (1.926 vs. 1.29) is expected: the training set captures a broader range of semantic variation, demanding a higher Information Premium to cover the full space of instructional constraints. Notably, V5's higher p-MRR (0.2360 vs. 0.2243) demonstrates that training-set derivation can actually *outperform* test-set grid search — eliminating overfitting risk while achieving better results.

This encoder-agnostic approach ensures that DeIR-Dual can be applied to any dense retriever without manual re-calibration: the parameters are computed from the retriever's own score distributions, not from external benchmarks.

5 Experiments

5.1 FollowIR Benchmark Background

Table 2 reports the performance of existing retrieval methods on the FollowIR benchmark [7]. Cross encoders (MonoT5-3B, Mistral-7B-instruct, FollowIR-7B) and bi-encoder dense models (BM25, TART-Contriever, Instructor XL, E5-Mistral, Google Gecko, RepLLaMA, Promptriever) are evaluated on three TREC collections. Most methods exhibit near-zero or negative p-MRR, indicating poor instruction sensitivity. Promptriever, which is fine-tuned on instruction-annotated data, achieves the best overall score but requires supervised training.

5.2 Main Results: FollowIR Benchmark

Table 3 compares DeIR-Dual with six query rewriting baselines on the FollowIR benchmark. DeIR-Dual achieves the highest mean p-MRR (14.9) and Score (29.0), outperforming all baselines by 10.6× in p-MRR (vs. HyDE) and 50.6× (vs. DeepRetrieval), while also achieving the highest retrieval quality.

5.3 Cross-Dataset Validation

[TODO] NegConstraint (text retrieval with negation signals); COCO-Neg (multimodal text-to-image negation). Both show significant improvement (+4.3% nDCG@10 on NegConstraint).

5.4 BEIR Generalization

[TODO] NQ (+1.1% nDCG@10), HotpotQA (-0.5%, within variance), MS MARCO (+2.0% nDCG@10). CONSERVATIVE prompting avoids pseudo-negation disaster.

	Model	Robust04 MAP	Robust04 p-MRR	News21 nDCG	News21 p-MRR	Core17 MAP	Core17 p-MRR	Average Score	Average p-MRR
Cross Encoders	MonoT5-3B	27.3	+4.0	16.5	+1.8	18.2	+1.8	20.7	+2.5
	Mistral-7B-instruct	23.2	+12.6	27.2	+4.8	19.7	+13.0	23.4	+10.1
	FollowIR-7B	24.8	+13.7	29.6	+6.3	20.0	+16.5	24.8	+12.2
Bi Encoders	BM25	12.1	-3.1	19.3	-2.1	8.1	-1.1	13.2	-2.1
	TART-Contriever	14.3	-9.0	21.8	-3.0	13.3	-3.0	16.5	-5.0
	Instructor XL	19.7	-8.1	26.1	-0.9	16.8	0.7	20.9	-2.8
	E5-Mistral	23.1	-9.6	27.8	-0.9	18.3	+0.1	23.1	-3.5
	Google Gecko	23.3	-2.4	29.5	+3.9	23.2	+5.4	25.3	+2.3
	RepLLaMA	24.0	-8.9	24.5	-1.8	20.6	+1.3	23.0	-3.1
	Promptriever	28.3	+11.7	28.5	+6.4	21.6	+15.4	26.1	+11.2
	DeIR-Dual	**28.6**	+8.3	**32.3**	**+18.7**	**26.0**	**+17.6**	**29.0**	**+14.9**

Table 2: Performance of existing retrieval methods and DeIR-Dual on the FollowIR benchmark [7]. All metrics are percentages ($\times$100). DeIR-Dual achieves the highest News21 nDCG, Core17 MAP, average score, and average p-MRR without any fine-tuning.

Method	Core17 MAP	Core17 p-MRR	Robust04 MAP	Robust04 p-MRR	News21 nDCG	News21 p-MRR	Average Score	Average p-MRR	Type
DeepRetrieval	21.5	+6.6	25.9	-5.3	23.9	-2.2	23.8	-0.3	RL query gen.
HyDE	23.6	+6.5	24.9	-2.9	21.4	+0.7	23.4	+1.4	Hypo. docs
Query2Doc	25.9	+8.0	27.9	-7.5	24.9	-3.8	26.2	-1.1	Pseudo-doc
RAG-Fusion	21.9	+5.4	21.1	-8.1	21.0	+1.8	21.3	-0.3	Multi-query
RAG-QR	22.3	+2.3	27.6	-10.3	22.9	-2.1	24.3	-3.4	T5 rewriter
DeIR-Dual	**26.0**	**+17.6**	**28.6**	**+8.3**	**32.3**	**+18.7**	**29.0**	**+14.9**	Dual-track

Table 3: Comparison of DeIR-Dual with six query rewriting methods on the FollowIR benchmark. DeIR-Dual achieves the highest mean p-MRR (14.9) and Score (29.0), outperforming all baselines by 10.6$\times$ in p-MRR (vs. HyDE) and 50.6$\times$ (vs. DeepRetrieval). All metrics are percentages ($\times$100).

5.5 Ablation Study

[TODO] Ablate each V2 component: dynamic τ, safety gate, gap weighting, Softplus penalty, Q_{minus} contribution.

5.6 Encoder-Agnostic Analysis

[TODO] EAPS strategy: RepLLaMA (0.08% at-risk) vs Mistral (62.9% at-risk). Retrieval-simulated distractor sampling with top-k=1000.

6 Analysis: Semantic Entanglement

6.1 Definition and Variants

[TODO] Structural entanglement (ComLQ: negation clauses share semantics with positive intent). Generative entanglement (NQ: LLM generates pseudo-negations overlapping with Q_{base}).

6.2 Diagnostic Analysis

[TODO] $\cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}})$ distribution: NegConstraint (orthogonal) vs ComLQ (0.705 mean) vs NQ (0.625 mean). S_{neg} higher on relevant documents $\rightarrow$ double whammy effect.

Figure 3: Semantic Entanglement Distribution
(placeholder: to be replaced with violin/box plot)

Figure 2: Distribution of $\cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}})$ across three scenarios. NegConstraint shows orthogonality (penalty effective), while ComLQ (structural entanglement) and NQ (generative entanglement) exhibit significant overlap, causing penalty failure.

7 Conclusion and Limitations

[TODO] Summary: three contributions. Limitations: semantic entanglement, LLM dependency, high at-risk encoder sensitivity. Future directions.

References

[1] Gianni Amati and Cornelis Joost Van Rijsbergen. Probabilistic models of information retrieval based on measuring the divergence from randomness. *ACM Transactions on Information Systems (TOIS)*, 2002.

[2] Akari Asai et al. Task-aware retrieval with instructions. In *arXiv preprint arXiv:2211.09260*, 2022.

[3] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. *arXiv preprint arXiv:2212.10496*, 2023.

[4] Dirk Groeneveld, Iz Beltagy, Pete Walsh, Akshita Bhagia, Rodney Kinney, et al. Olmo: Accelerating the science of language models. *arXiv preprint arXiv:2402.00838*, 2024.

[5] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *ICML*, 2017.

[6] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *EMNLP*, 2020.

[7] Gehan Kotonya et al. Followir: Evaluating and teaching instruction-following models for information retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.

[8] Xiang Li et al. Deepretrieval: Retrieval-augmented query generation for information retrieval. *arXiv preprint*, 2025.

[9] Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. Query rewriting for retrieval-augmented large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

[10] Niklas Muennighoff et al. Generative representational instruction tuning. *arXiv preprint arXiv:2402.09906*, 2024.

[11] Niklas Muennighoff, Nouamane Tazi, Loec Magne, and Nils Reimers. Mteb: Massive text embedding benchmark. In *EACL*, 2023.

[12] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *NeurIPS*, 2022.

[13] John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 1999.

[14] Simon Raudaschl. Rag-fusion: A new take on retrieval-augmented generation, 2024.

[15] Victor Sanh, Albert Webson, Colin Raffel, Stephen Bach, Linting Patry, et al. Multitask prompted training enables zero-shot task generalization. *ICLR*, 2022.

[16] Hong Su et al. One embedder, any task: Instruction-finetuned text embeddings. In *NeurIPS*, 2022.

[17] TBD. Coco-neg: Evaluating negation in multimodal retrieval. *arXiv preprint*, 2024.

[18] TBD. Deo: Disentangled explicit optimization for instruction-following retrieval. *arXiv preprint*, 2024. **[TODO] cite needed also known as NegConstraint dataset**.

[19] TBD. Deo: Training-free direct embedding optimization for negation-aware retrieval. *arXiv preprint arXiv:2603.09185*, 2026.

[20] TBD. Followtable: A benchmark for instruction-following table retrieval. *arXiv preprint arXiv:2605.00400*, 2026.

[21] Nandan Thakur, Nils Reimers, Andreas Rücklé, Iryna Gurevych, and Johannes Daxenberger. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *NeurIPS Datasets and Benchmarks*, 2021.

[22] Hugo Touvron, Thibaut Lavril, Gautier Izac, Marie-Anne Martinet, Amelie Lachaux, Timothee Lacroix, Baptiste Roziere, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.

[23] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, et al. Improving text embeddings with large language models. *arXiv preprint arXiv:2401.00368*, 2023.

[24] Zhicheng Wang, Yujia Zhang, Cheng Lu, Xiang Zhang, et al. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

[25] Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Yu, Brian Dunselman, Quoc V. Dai, and Ed H. Le. Finetuned language models are zero-shot learners. *ICLR*, 2022.

[26] Orion Weller, Benjamin Chang, Dawn Lawrie, Sean MacAvaney, Benjamin Van Durme, and Mahsa Yarmohammadi. Promptriever: Instruction-trained retrievers can be prompted like language models. *arXiv preprint arXiv:2409.11136*, 2024.

[27] Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. C-pack: Packaged resources to advance general chinese embedding. *arXiv preprint arXiv:2309.07597*, 2023.

[28] Mert Yuksekgonul et al. Negclip: Training clip to understand negation. In *arXiv preprint arXiv:2210.14697*, 2022.

[29] Qingcheng Zeng et al. Dual-view training for instruction-following information retrieval. *arXiv preprint arXiv:2604.18845*, 2026.

[30] Yuchen Zhuang, Aaron Trinh, Rushi Qiang, Haotian Sun, Chao Zhang, Hanjun Dai, and Bo Dai. Towards better instruction following retrieval models. *arXiv preprint arXiv:2505.21439*, 2025.